



Securing and Scaling the Data Pipeline

AWS Academy Data Engineering

Introduction

Securing and Scaling the Data Pipeline

Module objectives

This module prepares you to do the following:

- Highlight how cloud security best practices apply to analytics and machine learning (ML) data pipelines.
- List AWS services that play key roles in securing a data pipeline.
- Describe how infrastructure as code (IaC) supports the security and scalability of a data pipeline infrastructure.
- Identify the function of common AWS CloudFormation template sections.

Module overview

Presentation sections

- Cloud security review
- Security of analytics workloads
- ML security
- Scaling: An overview
- Creating a scalable infrastructure
- Creating scalable components

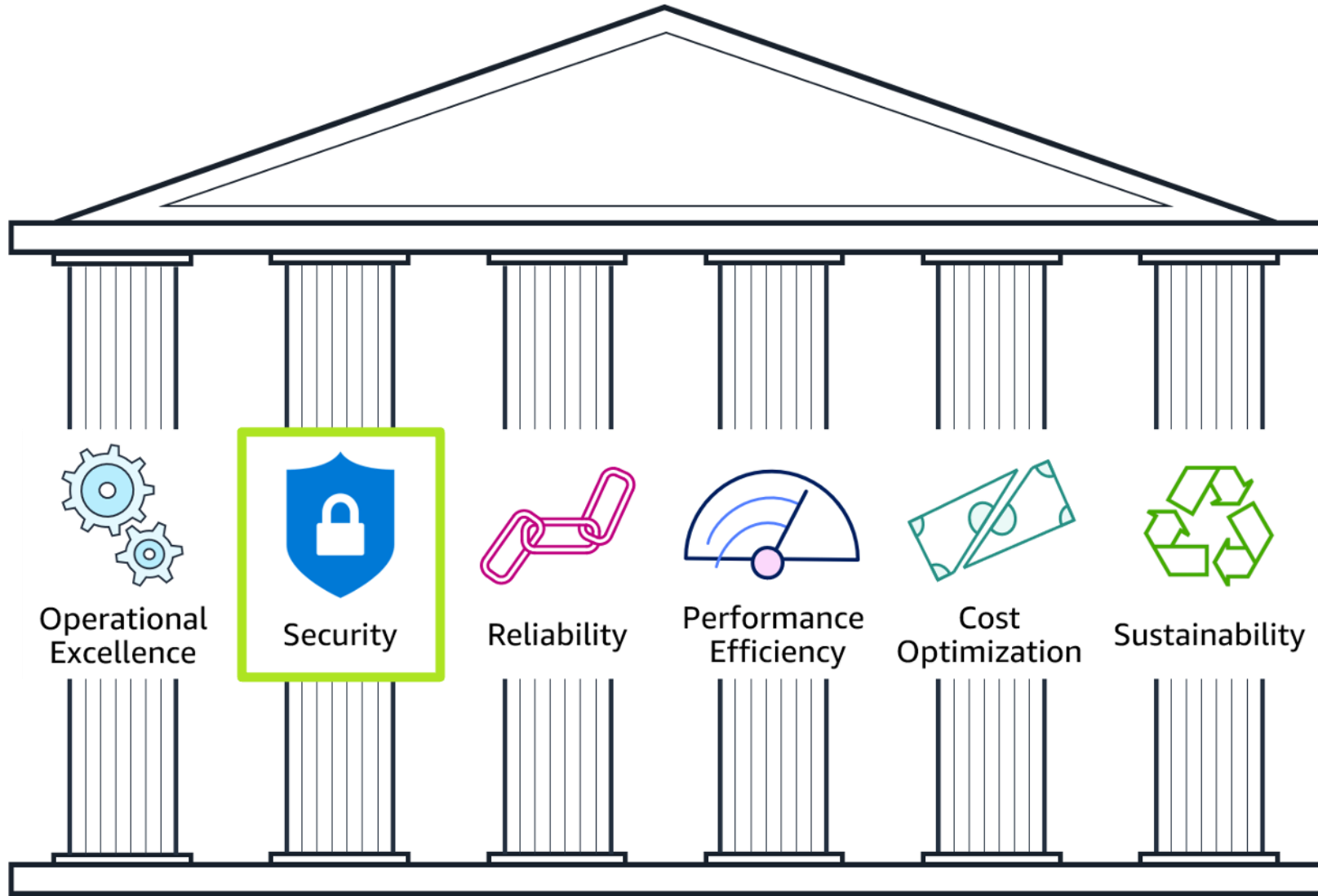
Knowledge checks

- Online knowledge check
- Sample exam question

Cloud security review

Securing and Scaling the Data Pipeline

AWS Well-Architected Framework: Security



Shared responsibility model

Customer

Responsibility for security
in the cloud

Customer data

Platform, applications, identity and access management

Operating system, network, and firewall configuration

Client-side data encryption
and data integrity,
authentication

Server-side encryption
(file system and data)

Networking traffic
protection (encryption,
integrity, identity)

AWS

Responsibility for security
of the cloud



AWS foundation services

Compute

Storage

Databases

Networking

AWS Global Infrastructure

Regions

Availability Zones

Edge Locations

Design principles for data security

- Implement a strong identity foundation.
- Enable traceability.
- Apply security at all layers.
- Automate security best practices.
- Protect data in transit and at rest.
- Keep people away from data.
- Prepare for security events.



Access management

Authentication

- Uses credentials to establish the identity of the requestor
- Grants or denies access to resources based on identity
- Utilizes usernames, passwords, and multi-factor authentication (MFA) among other methods

Authorization

- Takes place only after authentication
- Determines the level of access that an identity has to a resource
- Common methods include attribute-based access control (ABAC) and role-based access control (RBAC)

Principle of least privilege

- Grant only the permissions that are required to perform a task
- Start with a minimum set of permissions
- Grant additional permissions as necessary
- Revoke unnecessary permissions

AWS Identity and Access Management (IAM)

- Helps you to securely share and control access to your AWS resources for individuals and groups
- Integrates with most AWS services
- Supports federated identity management
- Supports granular permissions
- Supports MFA
- Provides identity information for information assurance and compliance audits



Data security

Data at rest

Any data that persists in nonvolatile storage for any duration

Protections

- Implement secure key management.
- Enforce encryption at rest.
- Enforce access control.
- Audit the use of encryption keys.
- Use mechanisms to keep people away from data.
- Automate data-at-rest protection.
- Audit data access logs.

Data in transit

Any data that is sent from one system to another

Protections

- Implement secure key and certificate management.
- Enforce encryption in transit.
- Authenticate network communications.
- Automate detection of unintended data access.
- Secure data from between VPC or on-premises locations.

AWS Key Management Service (AWS KMS)

- Provides the ability to create and manage cryptographic keys
- Uses hardware security modules (HSMs) to protect your keys
- Is integrated with other AWS services
- Provides the ability to set usage policies to determine which users can use which keys



Logging and monitoring

Logging

- Logging is the collection and recording of activity and event data.
- The information logged varies based on the service.
- Common log elements include date and time of event, origin of event, and identity of resources that were accessed.

Monitoring

- Monitoring is the continuous verification of the security and performance of your resources, applications, and data.
- AWS provides several services that give you the visibility to spot issues before they impact operations.

AWS CloudTrail

- Is the primary AWS solution for logging
- Assists you to enable governance and compliance as well as operational and risk auditing of your AWS account
- Records actions taken by a user, role, or AWS service as events
- Can be used to view, search, download, archive, analyze, and respond to account activity across your AWS infrastructure



Amazon CloudWatch

- Is a monitoring and observability service
- Provides a unified view of the operational health of your AWS resources, applications, and services
- Collects metrics in the AWS Cloud and on premises
- Can be used to monitor and troubleshoot infrastructure
- Provides the ability to customize logs and events



Key takeaways: Cloud security review



- Access management consists of authentication and authorization; adhere to the principle of least privilege with both.
- IAM integrates with most AWS services and helps you to securely share and control individual and group access to your AWS resources.
- A key aspect of a data security plan is to secure data at rest and data in transit.
- Logging and monitoring can assist your organization to maintain compliance with local laws and regulations.

Security of analytics workloads

Securing and Scaling the Data Pipeline

Classify and protect data



Understand data
classifications and policies



Identify the source data
owners



Record data classifications
into the Data Catalog



Implement data encryption
policies



Implement data retention
policies



Honor classifications
downstream

— = required

- - = recommended

Control the data access



Allow data owners to
determine access



Build user identity
solutions



Implement data access
authorization models



Establish an emergency
access process

 = required

 = recommended

Control the access to workload infrastructure



Prevent unintended
access



Implement least privilege
policies



Monitor the
infrastructure changes
and user activities

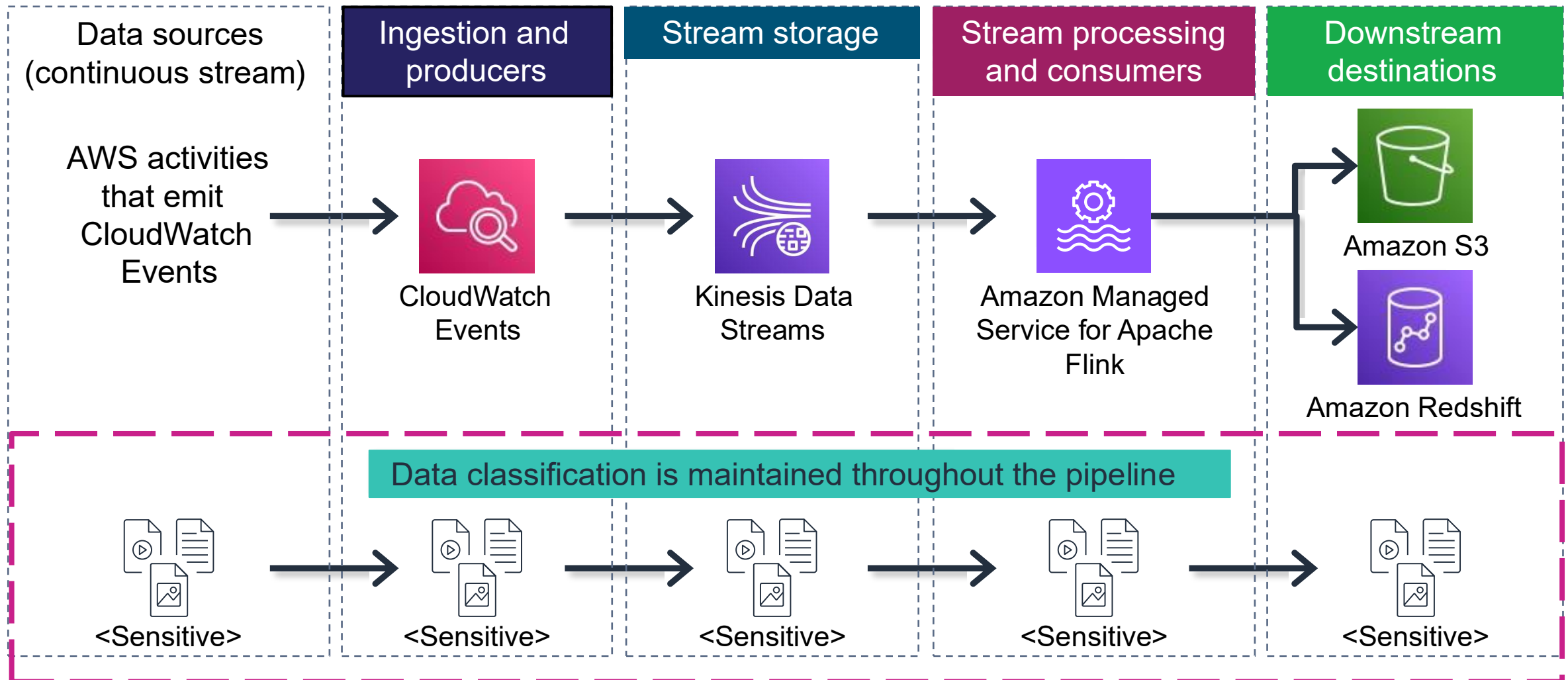


Secure infrastructure
audit logs

 = required

 = recommended

Securing the stream processing pipeline



Key takeaways: Security of analytics workloads

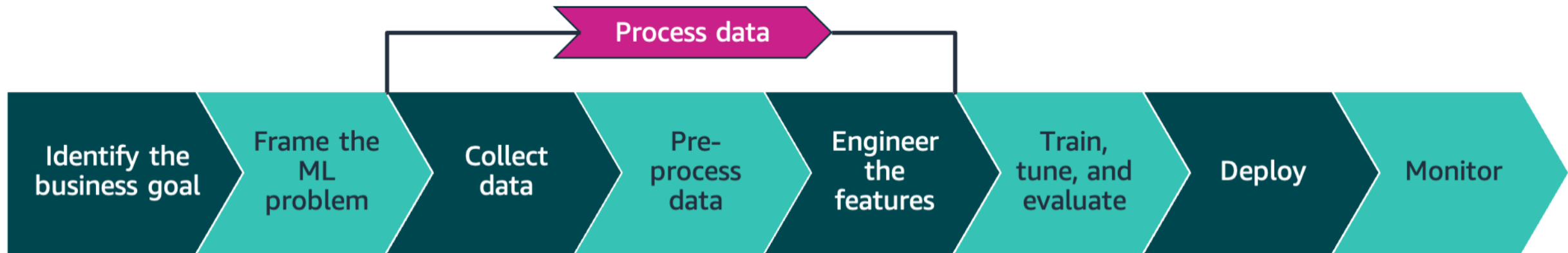


- Honor the data classifications and protection policies that the owners of the source data assigned.
- Secure access to the data in the analytics workload.
- Share data downstream in compliance with the source system's classification policies.
- Ensure the environment is accessible with the least permissions necessary; automate auditing of environment changes, and alert in case of abnormal environment access.

ML security

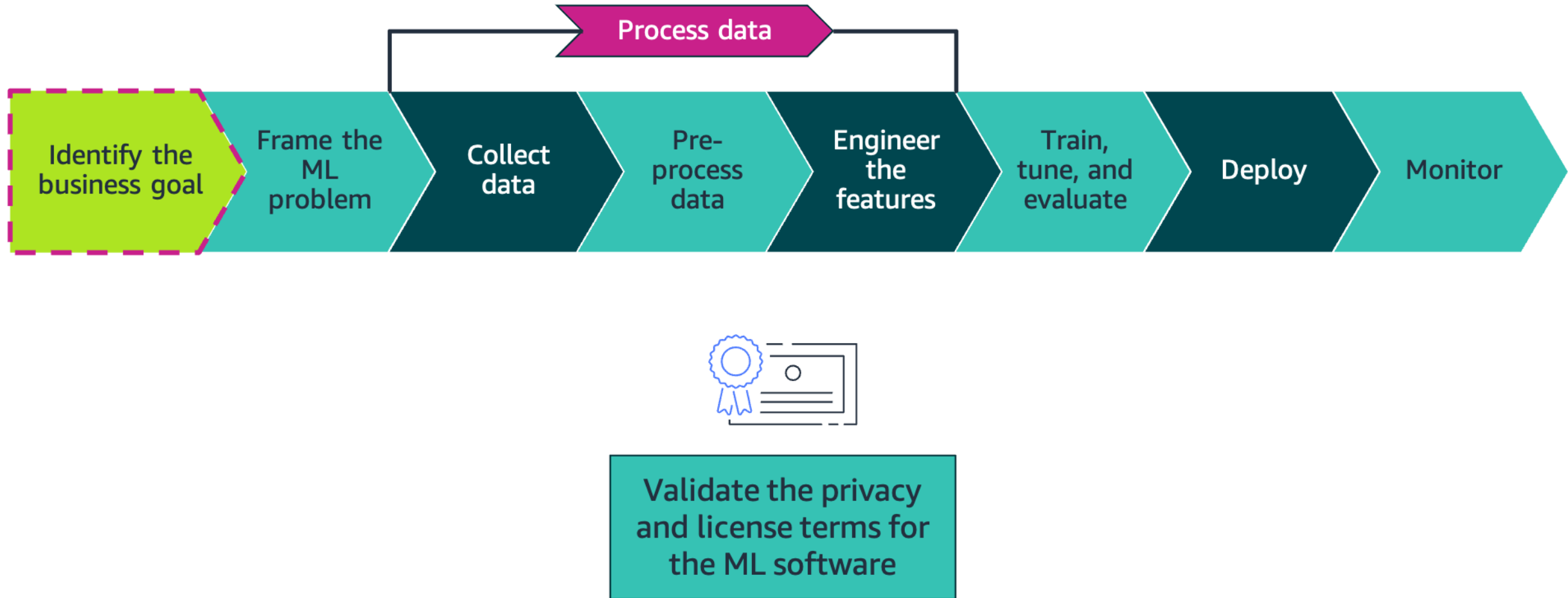
Securing and Scaling the Data Pipeline

AWS Well-Architected Framework: ML Lens

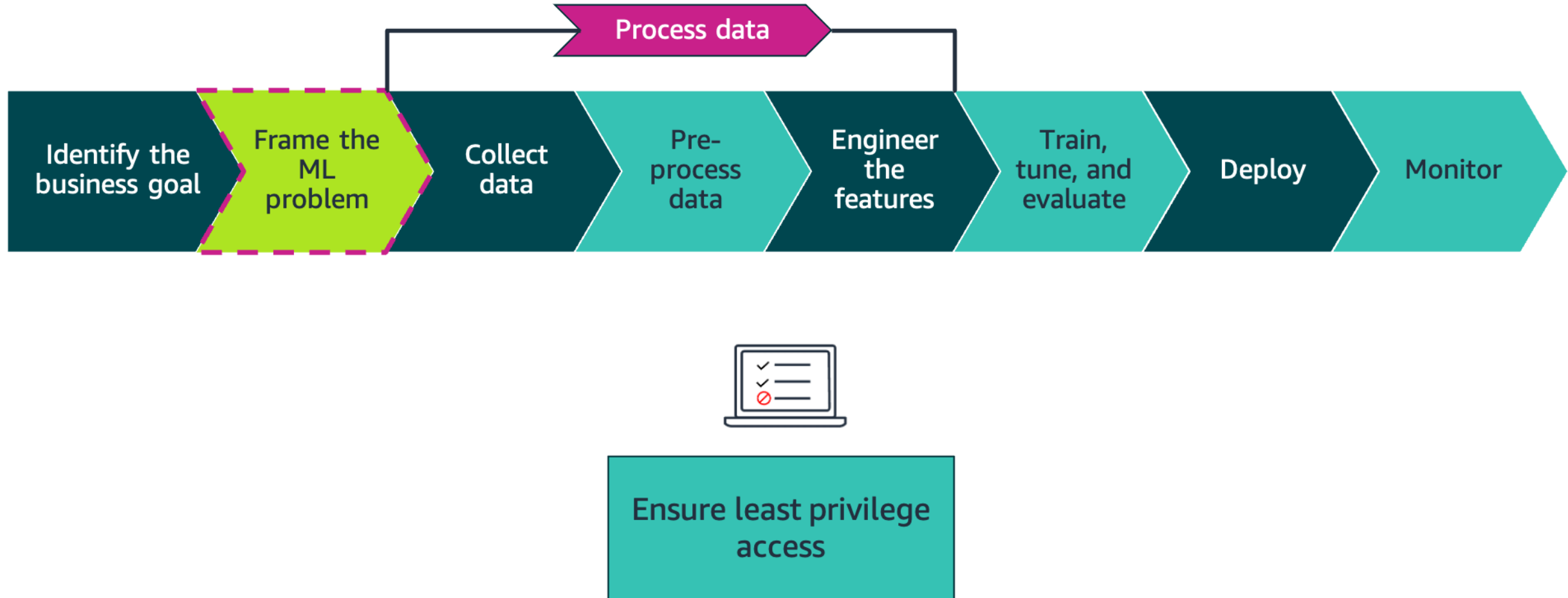


ML lifecycle

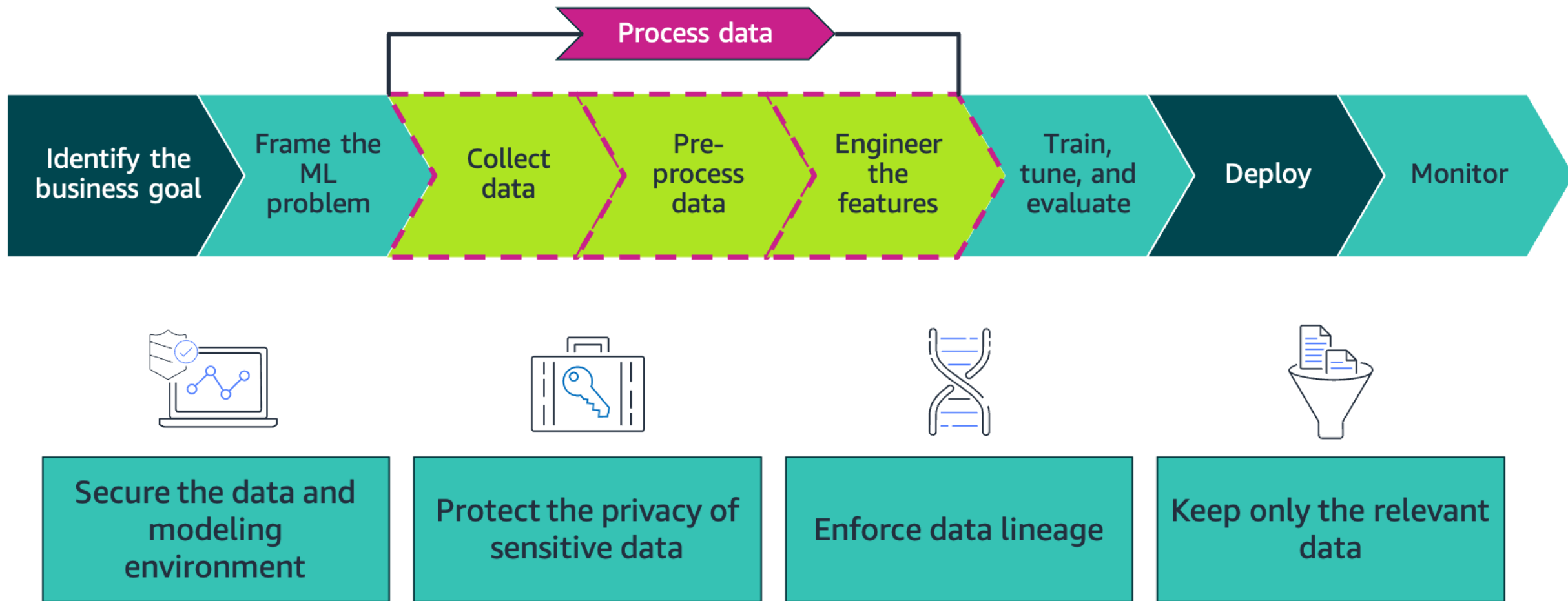
Phase: Identify the business goal



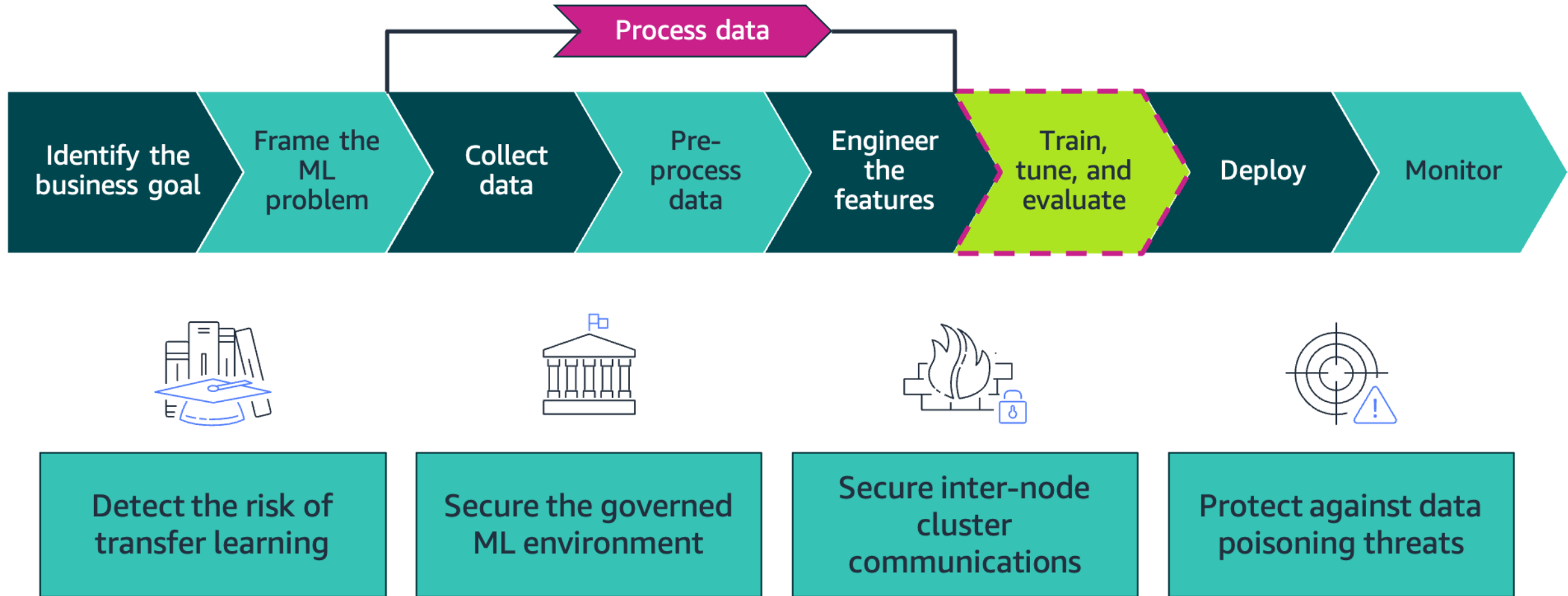
Phase: Frame the ML problem



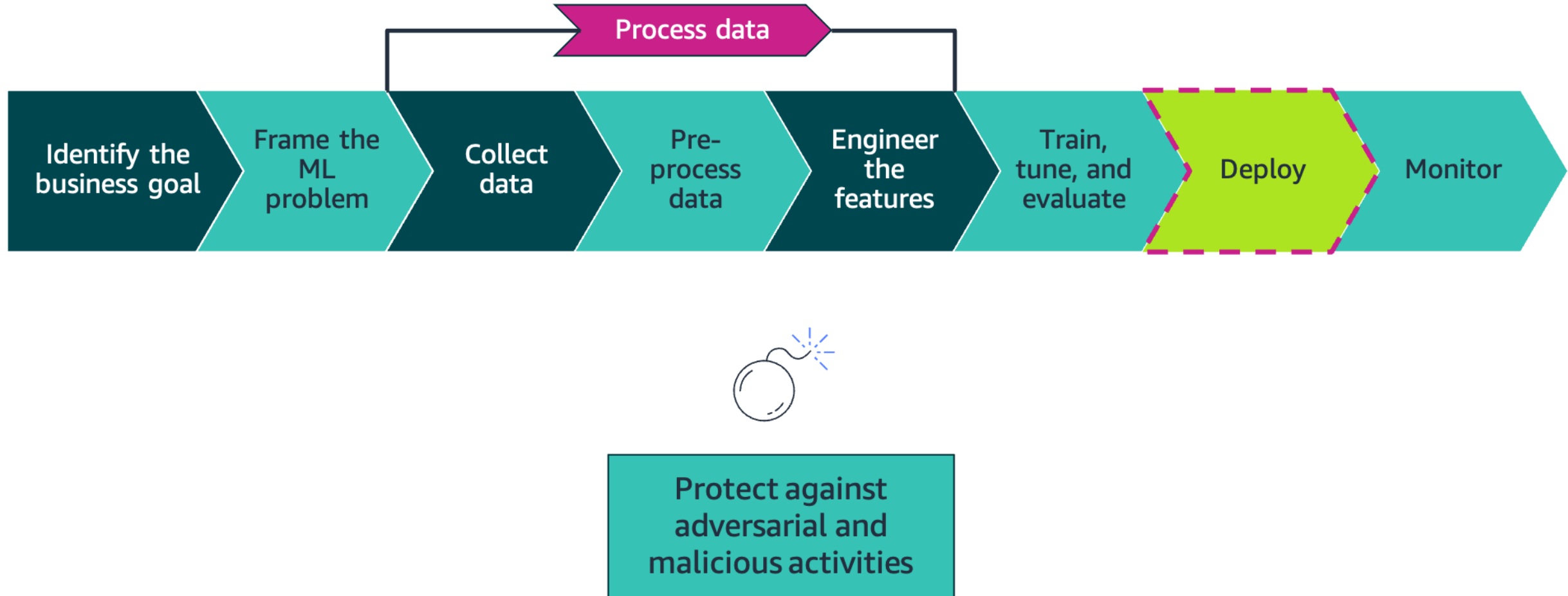
Phase: Process data



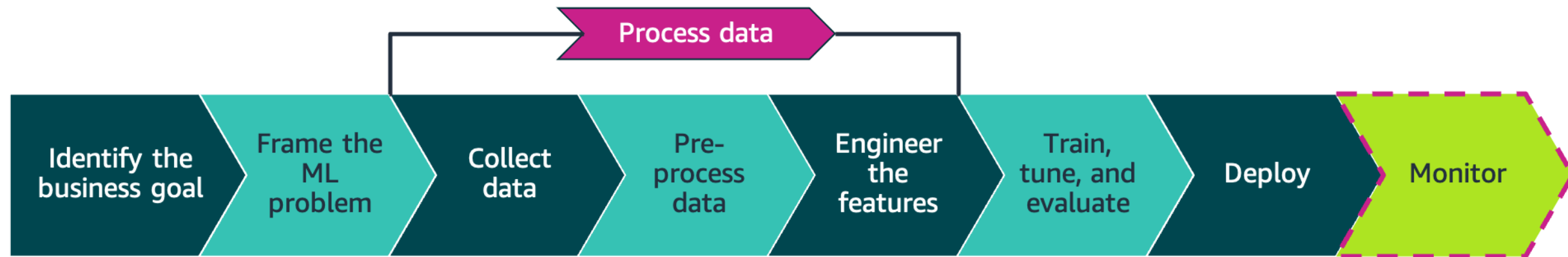
Phase: Train, tune, and evaluate



Phase: Deploy



Phase: Monitor



Restrict access to
intended legitimate
customers



Monitor human
interactions with data
for anomalous activity

Key takeaways: ML security



- Apply the principle of least privilege throughout the ML lifecycle.
- Encrypt data in transit to and at rest in the compute and storage infrastructure.
- To reduce data exposure risks, only store the data that has business need.
- To detect malicious inputs that might result in incorrect predictions, add protection inside and outside of the deployed code.
- Ensure that data access logging is enabled, and audit for anomalous data access events.

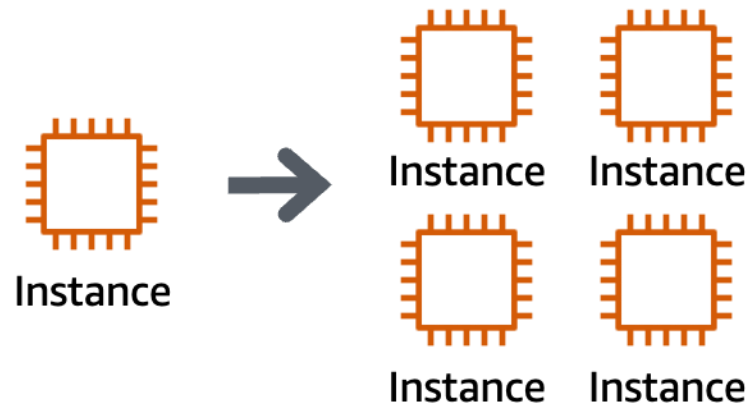
Scaling: An overview

Securing and Scaling the Data Pipeline

Types of scaling

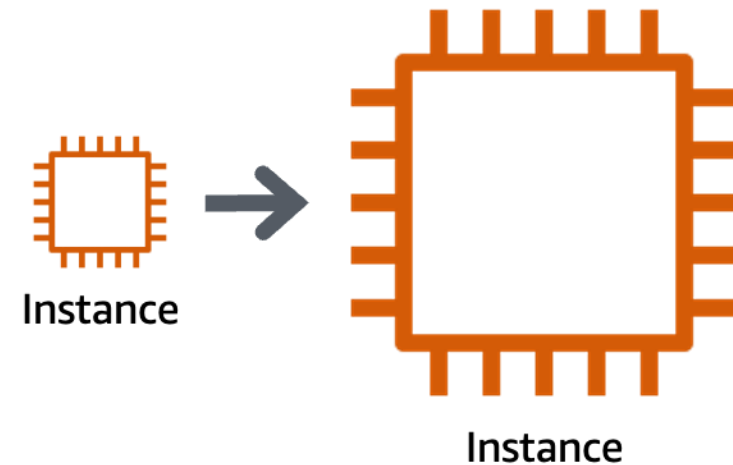
Horizontal

Increases capacity by adding more computers to the system



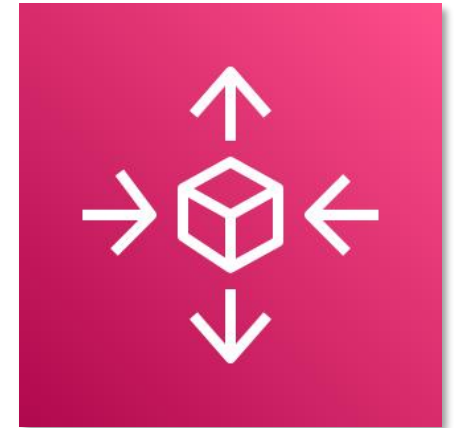
Vertical

Increases performance by adding more resources into one computer



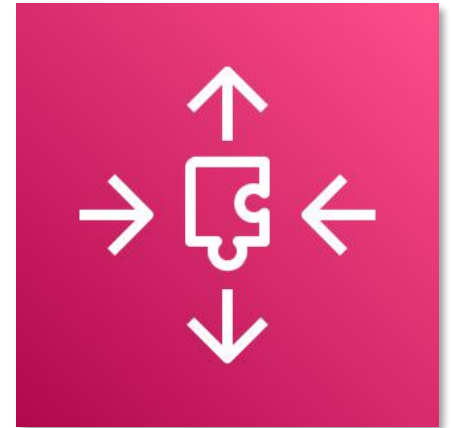
AWS Auto Scaling

- Automatically adjusts capacity
- Set target resource utilization levels
- Create scaling plans
- Continuous performance monitoring
- Cost efficient
- Predictive scaling



AWS Application Auto Scaling

- Automatically scale scalable services and resources
- Conditions-based
- Target tracking scaling
- Step scaling
- Scheduled scaling

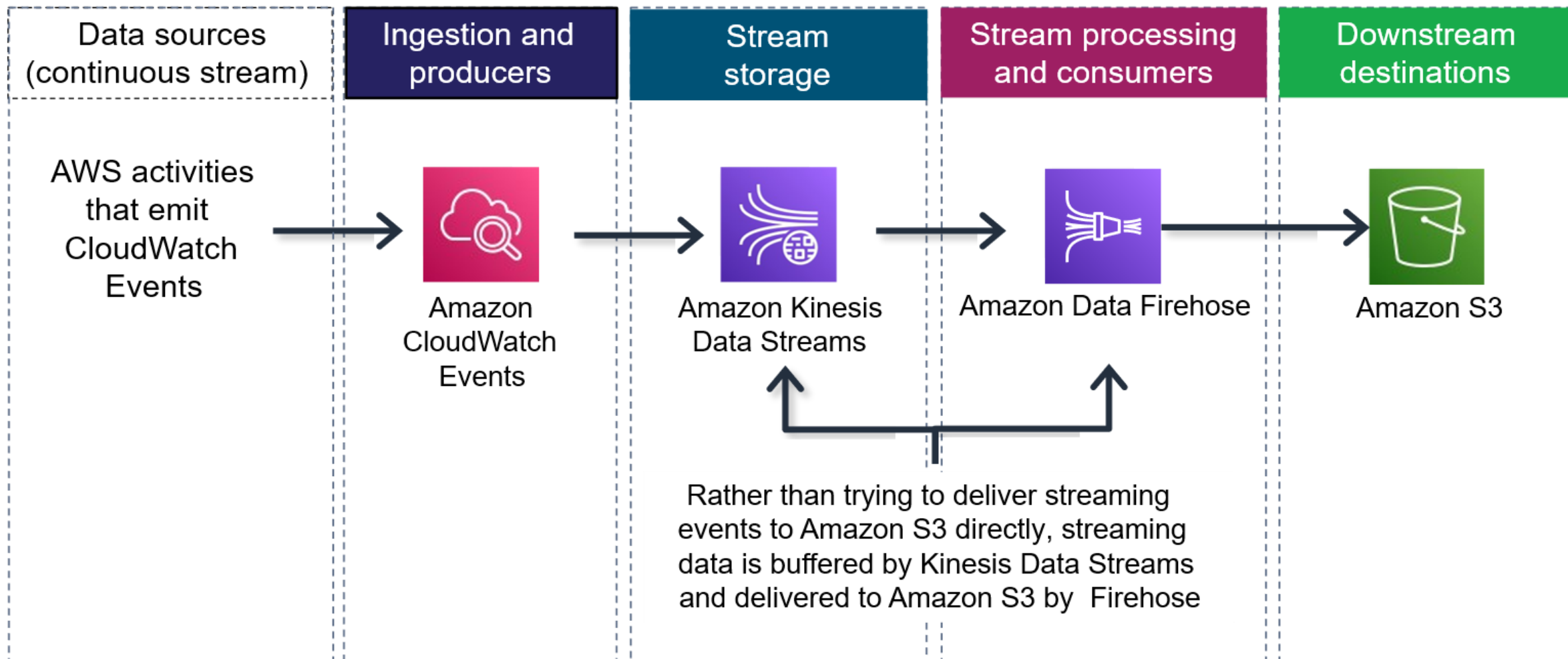


Scaling and cost management

- Identify the key performance goals for your workload.
- Design your workload to allow for components to scale independently
- Use metrics to find and address bottlenecks and issues.
- Make trade-offs that align to your key performance goals.



Example: Scaling in a stream processing pipeline



Key takeaways:

Scaling: An overview



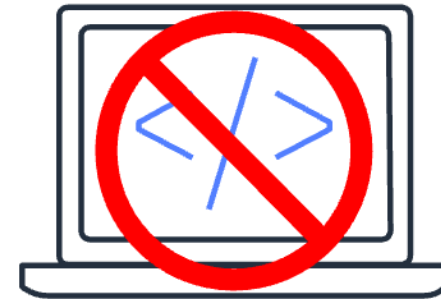
- Horizontal scaling adds additional instances; vertical scaling adds additional resources to an instance.
- AWS Auto Scaling automatically scales Amazon EC2 capacity horizontally or vertically based on scaling plans and predictive scaling.
- Application Auto Scaling automatically scales resources for individual AWS services beyond Amazon EC2.

Creating a scalable infrastructure

Securing and Scaling the Data Pipeline

Issues with traditional infrastructure deployments

- Is time-consuming
- Has tasks that can be repetitive and prone to error
- Is a combination of scripts and manual processes
- Inadequate version control of scripts and processes
- Can lead to nonrepeatable, unreliable, or inconsistent deployments



Infrastructure as code

- Automates the process to create, update, and delete AWS infrastructure
- Stands up identical dev/test environments on demand
- Uses the same code to create your production environment that you used to create your other environments
- Provides the ability to create, deploy, and maintain infrastructure in a programmatic, descriptive, and declarative manner
- Benefits from the rigor, clarity, and reliability provided by CloudFormation and the AWS Cloud Development Kit (AWS CDK)

Infrastructure as code methods

Infrastructure as code can use a *declarative* or *imperative* programming approach

Declarative

- Defines the resources
- Defines the resource properties
- Defines the system state
- Defines the tools that are used to configure your environment

Defines what, not how

Imperative

- Provides the equivalent of step-by-step instructions
- Focuses on the desired end state

Defines what *and* how, step by step

AWS CloudFormation

- Is a fully managed service
- Creates, updates, and deletes resources in stacks
- Automates AWS resource provisioning
- Simplifies the task of repeatedly and predictably creating groups of related resources that power your applications



How CloudFormation works

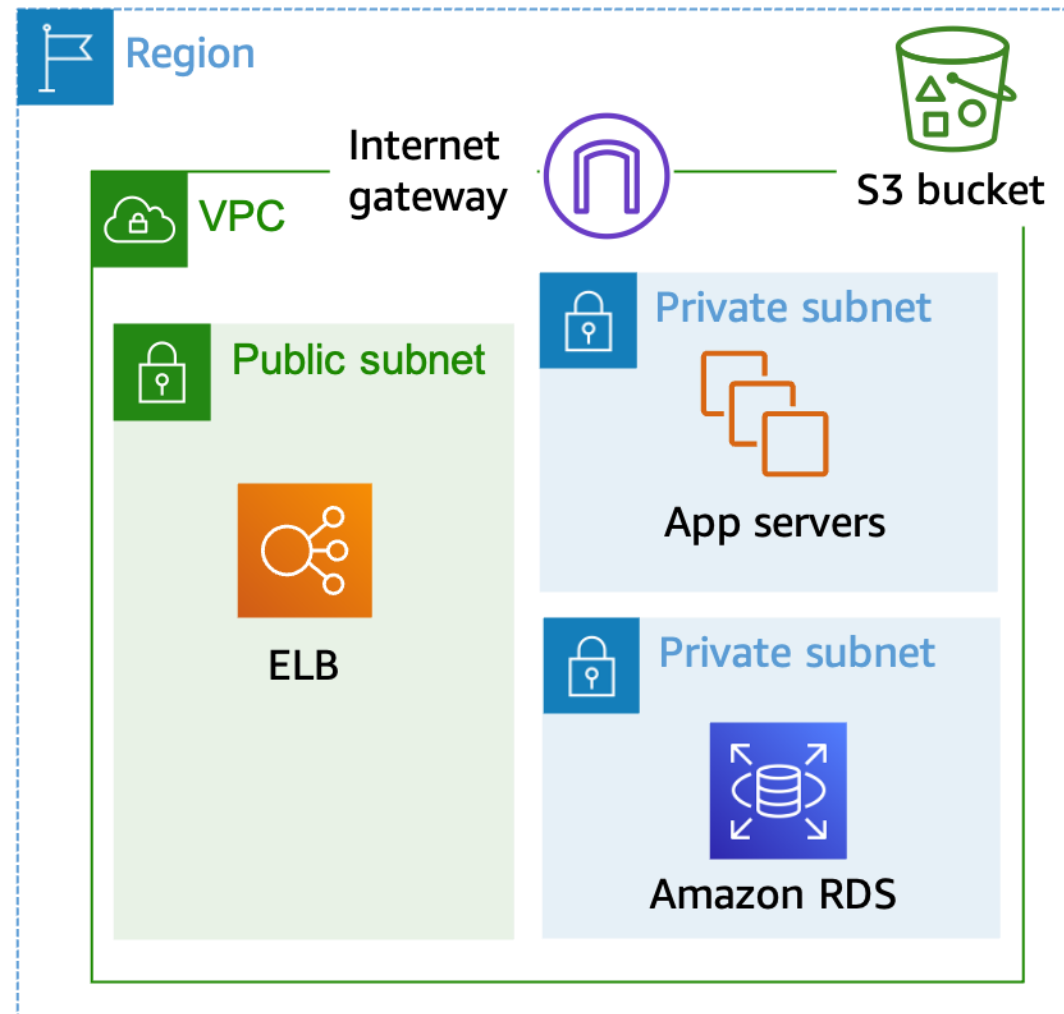
```
"Ec2Instance" : {  
  "Type" : "AWS::EC2...",  
  "Properties" : {  
    "KeyName" : "My...",  
    "ImageId" : "ami...",  
    "InstanceType" :  

```

Template file



CloudFormation



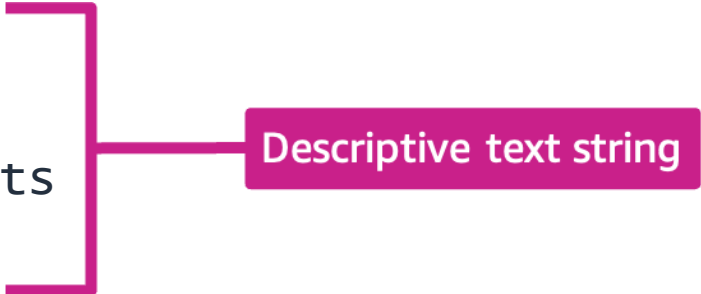
Stack

CloudFormation template structure

```
{  
  "AWSTemplateFormatVersion" : "version date",  
  "Description" : "JSON string",  
  "Metadata" : {template metadata},  
  "Parameters" : {set of parameters},  
  "Mappings" : {set of mappings},  
  "Conditions" : {set of conditions},  
  "Transform" : {set of transforms},  
  "Resources" : {set of resources},  
  "Outputs" : {set of outputs}  
}
```

CloudFormation template example: Description

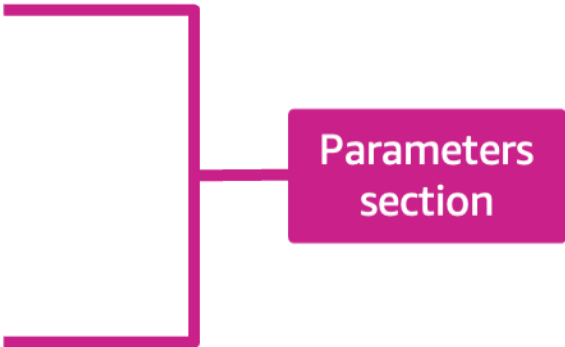
```
{  
  "AWSTemplateFormatVersion": "2010-09-09",  
  
  "Description": "AWS CloudFormation Sample Template  
Amazon S3 Pipeline: Configure and launch an  
Infrastructure that includes a pipeline that connects  
to an Amazon S3 source bucket."  
  
  "Parameters": {  
    "SourceObjectKey": {  
      "Description": "S3 source artifact",  
      "Type": "String",  
      "Default": "SampleApp_Linux.zip"  
    },  
  },  
}
```



Descriptive text string

CloudFormation template example: Parameters

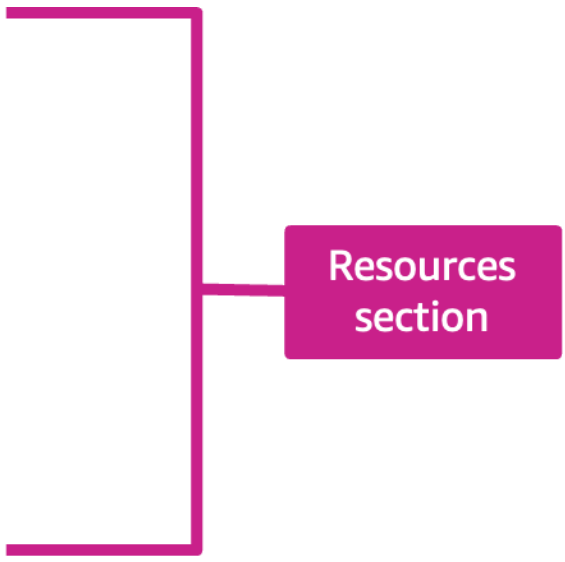
```
{  
  "AWSTemplateFormatVersion": "2010-09-09",  
  
  "Description": "AWS CloudFormation Sample Template  
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    "SourceObjectKey": {  
      "Description": "S3 source artifact",  
      "Type": "String",  
      "Default": "SampleApp_Linux.zip"  
    },  
  },  
}
```



Parameters
section

CloudFormation template example: Resources

```
"AWSTemplateFormatVersion": "2010-09-09",
"Resources": {
  "AppPipeline": {
    "Type": "AWS::CodePipeline::Pipeline",
    "Properties": {
      "Name": "s3-events-pipeline",
      "RoleArn": {
        "Fn::GetAtt": [
          "CodePipelineServiceRole",
          "Arn"
        ]
      }
    }
  },
}
```



Resources
section

AWS Cloud Development Kit (AWS CDK)

- AWS CDK apps are written in TypeScript, JavaScript, Python, Java, C#, or Go.
- AWS CDK apps use the AWS CDK to define AWS infrastructure.
- An app defines one or more stacks that each contain constructs that define concrete AWS resources.
- The AWS CDK includes the CDK Toolkit—a command line tool to work with AWS CDK apps and stacks.



Key takeaways: Creating a scalable infrastructure

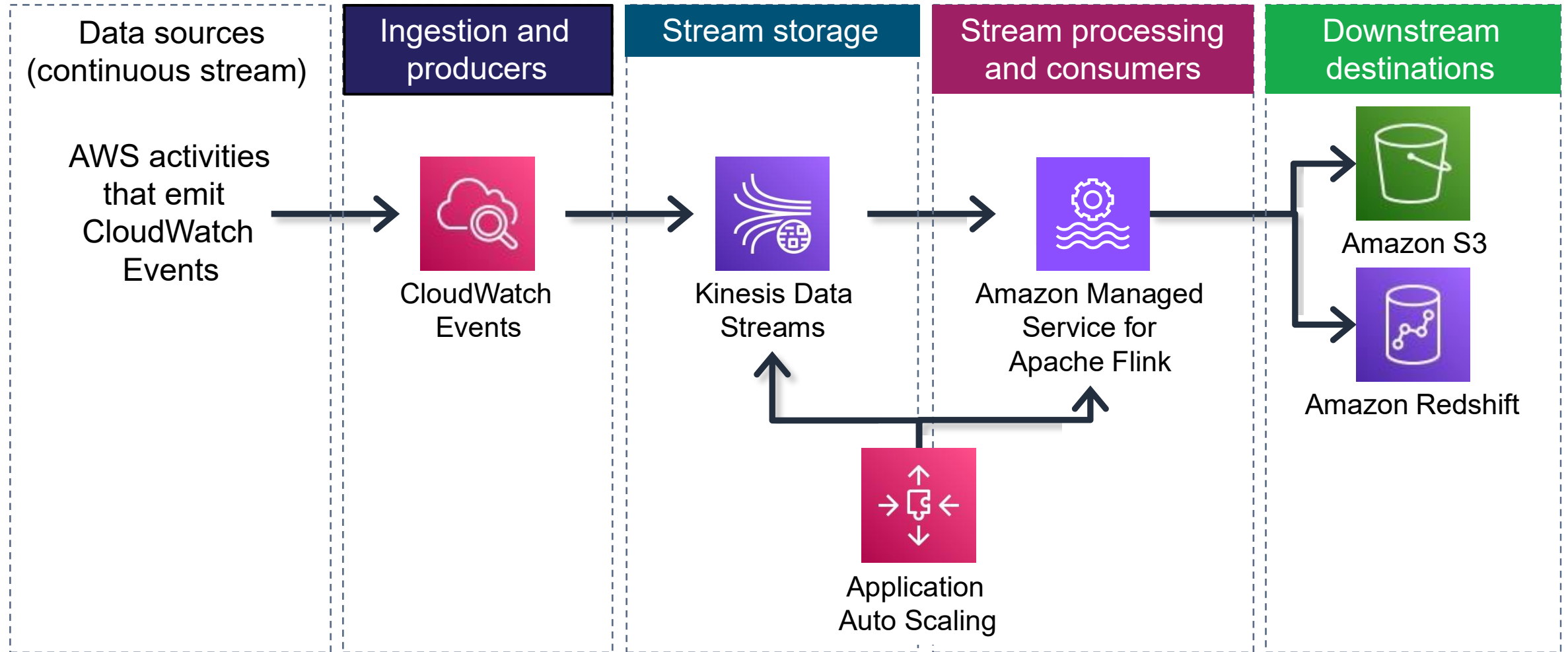


- Traditional infrastructure deployments use manual processes that are time-consuming and prone to error.
- IaC automates the process to create, update, and delete AWS infrastructure.
- IaC can use a declarative or imperative programming approach.
- You can use CloudFormation to automate AWS resource and pipeline provisioning.

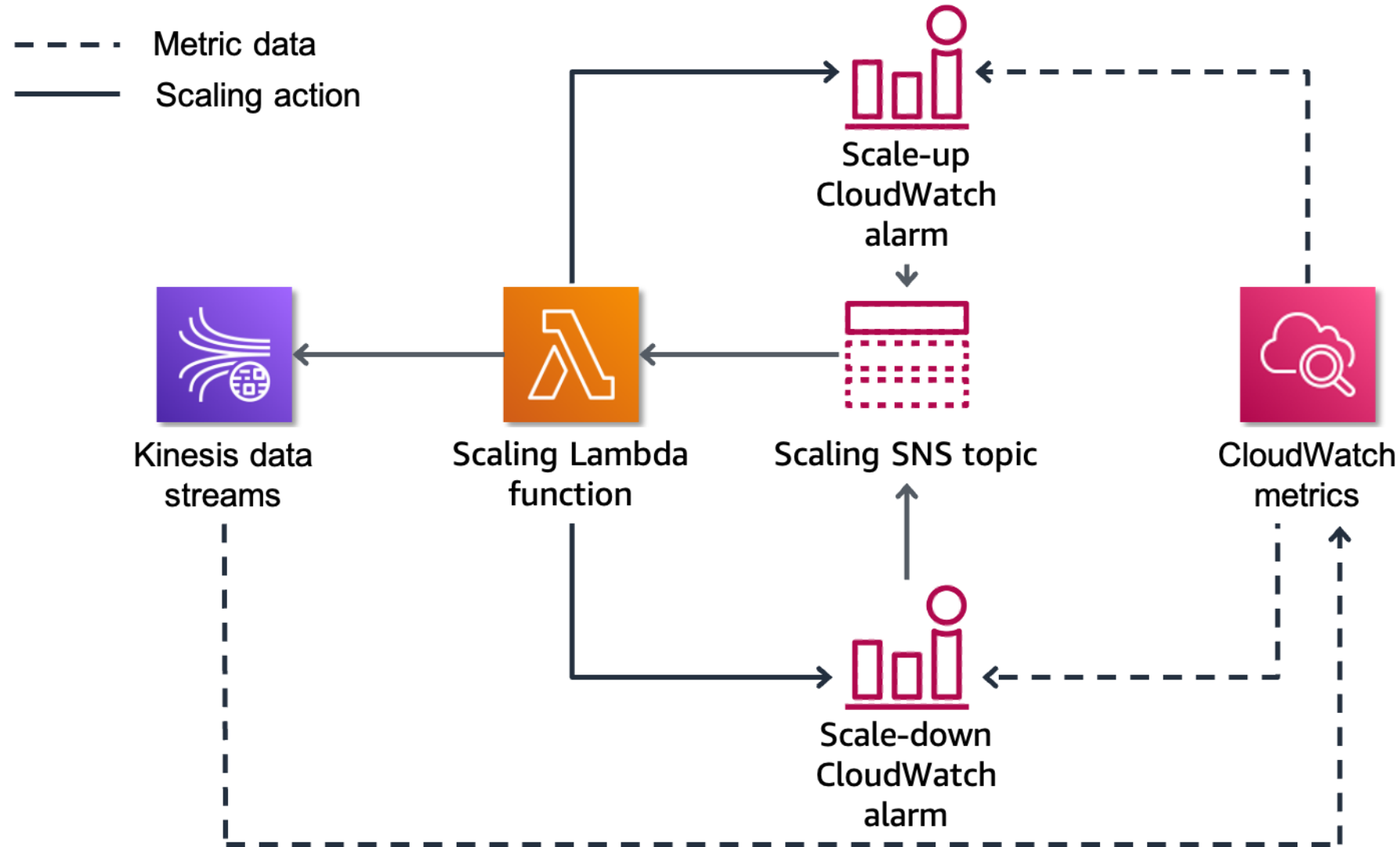
Creating scalable components

Securing and Scaling the Data Pipeline

Stream processing pipeline



Scaling Kinesis data streams



Key takeaways: Creating scalable components



- Configure scalable components in your pipeline to ensure that it is optimized.
- Use CloudWatch and Lambda to provide automatic scaling for Kinesis data streams.
- Kinesis Data Streams on-demand mode is the recommended method for automatic scaling.

Module wrap-up

Securing and Scaling the Data Pipeline

Module summary

This module prepared you to do the following:

- Highlight how cloud security best practices apply to analytics and ML data pipelines.
- List AWS services that play key roles in securing a data pipeline.
- Describe how IaC supports the security and scalability of a data pipeline infrastructure.
- Identify the function of common CloudFormation template sections.

Module knowledge check



- The knowledge check is delivered online within your course.
- The knowledge check includes 10 questions based on material presented on the slides and in the slide notes.
- You can retake the knowledge check as many times as you like.

Sample exam question

A data engineer is preparing to deploy a data lake that users and programs throughout the organization will access. The engineer wants to ensure that the Amazon EMR clusters that will support the architecture can maintain a consistent performance standard regardless of the demand.

Which service could the data engineer use to accomplish this task?

Identify the key words and phrases before continuing.

The following are the key words and phrases:

- **EMR clusters**
- **Consistent performance**

Sample exam question: Response choices

A data engineer is preparing to deploy a data lake that users and programs throughout the organization will access. The engineer wants to ensure that the Amazon **EMR clusters** that will support the architecture can maintain a **consistent performance** standard regardless of the demand.

Which service could the data engineer use to accomplish this task?

Choice	Response
A	AWS EC2 Auto Scaling
B	AWS Identity and Access Management (IAM)
C	AWS Application Auto Scaling
D	AWS CloudTrail

Sample exam question: Answer

The correct answer is C.

Choice	Response
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C	AWS Application Auto Scaling
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Thank you



Corrections, feedback, or other questions?

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